

Single Level Directory

```
#include <stdio.h>
#include <string.h>
int main() {
    int n, i, j, duplicate;
    char files[20][30];
    printf("Single Level Directory\n");
    printf("Enter number of files: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        while(1) {
            duplicate = 0;
            printf("Enter file %d name: ", i + 1);
            scanf("%s", files[i]);
            for(j = 0; j < i; j++) {
                if(strcmp(files[i], files[j]) == 0) {
                    duplicate = 1;
                    printf("Error: File name already exists.\n");
                    printf("Please enter a different file name.\n\n");
                    break;
                }
            }
            if(duplicate == 0) {
                break;
            }
        }
    }
    printf("\nFiles in the directory are:\n");
    for(i = 0; i < n; i++) {
        printf("%s\n", files[i]);
    }
    return 0;
}
```

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1WA24CS218
Single Level Directory
Enter number of files: 3
Enter file 1 name: abcd.txt
Enter file 2 name: pqrs.c
Enter file 3 name: abcd.jpg

Files in the directory are:
abcd.txt
pqrs.c
abcd.jpg

...Program finished with exit code 0
Press ENTER to exit console.
```

Two Level Directory

```
#include <stdio.h>
#include <string.h>
int main() {
    int users, files[20];
    int i, j, k, duplicate;
    char user[20][30];
    char filename[20][20][30];
    printf("Two Level Directory\n");
    printf("Enter number of users: ");
    scanf("%d", &users);
    for(i = 0; i < users; i++) {
        while(1) {
            duplicate = 0;
            printf("\nEnter username: ");
            scanf("%s", user[i]);
            for(k = 0; k < i; k++) {
                if(strcmp(user[i], user[k]) == 0) {
                    duplicate = 1;
                    printf("Error: Username already exists.\n");
                    printf("Enter a different username.\n\n");
                    break;
                }
            }
        }
    }
}
```

```

    }
}
if(duplicate == 0) {
    break;
}
}
printf("Enter number of files for %s: ", user[i]);
scanf("%d", &files[i]);
for(j = 0; j < files[i]; j++) {
    while(1) {
        duplicate = 0;
        printf("Enter file %d name: ", j + 1);
        scanf("%s", filename[i][j]);
        for(k = 0; k < j; k++) {
            if(strcmp(filename[i][j], filename[i][k]) == 0) {
                duplicate = 1;
                printf("Error: File already exists for this user.\n");
                printf("Enter a different filename.\n\n");
                break;
            }
        }
        if(duplicate == 0) {
            break;
        }
    }
}
}
printf("\n\nDirectory Structure:\n");
for(i = 0; i < users; i++) {
    printf("\nUser: %s\n", user[i]);
    for(j = 0; j < files[i]; j++) {
        printf("%s\n", filename[i][j]);
    }
}
return 0;
}

```

```
1WA24CS218
Two Level Directory
Enter number of users: 2

Enter username: Documents
Enter number of files for Documents: 2
Enter file 1 name: Project.txt
Enter file 2 name: notes.txt

Enter username: College
Enter number of files for College: 2
Enter file 1 name: Project.txt
Enter file 2 name: Image.jpg

Directory Structure:

User: Documents
Project.txt
notes.txt

User: College
Project.txt
Image.jpg

...Program finished with exit code 0
Press ENTER to exit console.
```

Hierarchical Directory

```
#include <stdio.h>
#include <string.h>
int main() {
    int mainDir, subDir[20], files[20][20];
    int i, j, k, l;
    int duplicate;
    char mainDirectory[20][30];
    char subDirectory[20][20][30];
```

```

char filename[20][20][20][30];
printf("Hierarchical Directory\n");
printf("Enter number of main directories: ");
scanf("%d", &mainDir);
for(i = 0; i < mainDir; i++) {
    while(1) {
        duplicate = 0;
        printf("\nEnter main directory name: ");
        scanf("%s", mainDirectory[i]);
        for(l = 0; l < i; l++) {
            if(strcmp(mainDirectory[i], mainDirectory[l]) == 0) {
                duplicate = 1;
                printf("Error: Main directory already exists.\n");
                printf("Enter a different directory name.\n\n");
                break;
            }
        }
        if(duplicate == 0) {
            break;
        }
    }
    printf("Enter number of subdirectories in %s: ", mainDirectory[i]);
    scanf("%d", &subDir[i]);
    for(j = 0; j < subDir[i]; j++) {
        while(1) {
            duplicate = 0;
            printf("Enter subdirectory name: ");
            scanf("%s", subDirectory[i][j]);
            for(l = 0; l < j; l++) {
                if(strcmp(subDirectory[i][j], subDirectory[i][l]) == 0) {
                    duplicate = 1;
                    printf("Error: Subdirectory already exists.\n");
                    printf("Enter a different subdirectory name.\n\n");
                    break;
                }
            }
            if(duplicate == 0) {
                break;
            }
        }
    }
}

```

```

printf("Enter number of files in %s: ", subDirectory[i][j]);
scanf("%d", &files[i][j]);
for(k = 0; k < files[i][j]; k++) {
    while(1) {
        duplicate = 0;
        printf("Enter file %d name: ", k + 1);
        scanf("%s", filename[i][j][k]);
        for(l = 0; l < k; l++) {
            if(strcmp(filename[i][j][k], filename[i][j][l]) == 0) {
                duplicate = 1;
                printf("Error: File already exists.\n");
                printf("Enter a different filename.\n\n");
                break;
            }
        }
        if(duplicate == 0) {
            break;
        }
    }
}
}
}
printf("\n\nHierarchical Directory Structure:\n");
for(i = 0; i < mainDir; i++) {
    printf("\n%s\n", mainDirectory[i]);
    for(j = 0; j < subDir[i]; j++) {
        printf("  %s\n", subDirectory[i][j]);
        for(k = 0; k < files[i][j]; k++) {
            printf("    %s\n", filename[i][j][k]);
        }
    }
}
return 0;
}

```

```
1WA24CS218
Hierarchical Directory
Enter number of main directories: 1

Enter main directory name: Documents
Enter number of subdirectories in Documents: 2
Enter subdirectory name: Project
Enter number of files in Project: 2
Enter file 1 name: code.c
Enter file 2 name: report.txt
Enter subdirectory name: Notes
Enter number of files in Notes: 2
Enter file 1 name: os.txt
Enter file 2 name: dbms.pdf
```

Hierarchical Directory Structure:

```
Documents
  Project
    code.c
    report.txt
  Notes
    os.txt
    dbms.pdf
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```